ESCAPEE GLOSS: A SYMPHONY OF POLYMEDIA

Brigid Burke
Independent
Composer/Performer/Artist
www.brigid.com.au

ABSTRACT

Escapee Gloss: A Symphony of Polymedia explores the methods and meaning of polymedia processes through the composition and subsequent performance of an interactive seven-movement piece, *Escapee Gloss* composed from 2010-14. The creative work is a documentation of musical and graphic scores and polymedia recordings, including my own performance as a clarinettist, electronic music artist and polymedia artist.

The work *Escapee Gloss*, embodies a response to some old, used clarinets, irreparable and unplayable, and sundry discarded clarinet parts. It integrates these physically and emotionally into experiments with sound, art works, layering of light, reflections and manipulated artwork and music in performance through multi-channel audio and video mixing. The composition combines digitally processed clarinet sounds, live acoustic instruments (flute, Eb/Bb and bass clarinets, double bass and piano), live processed acoustic environmental sounds, props and projections. These components reinvent the disintegrated clarinets through video samples and stills of broken clarinet parts, grey pencil drawings, water footage and Japanese artwork. *Escapee Gloss* explores the possibilities of layering and reflections of light and texture in images and sound and deliberately referencing paint textures in the titles of the movements.

The polymedia processes and multi-art forms cohere in live performance through a deliberate strategy of layering to represent the complexity and depth of the images we see and of the sounds we hear.

1. INTRODUCTION

The initial idea for *Escapee Gloss* came to me when I was given 32 old, used clarinets, irreparable and unplayable, from Kilvington Grammar School in Ormond, Melbourne. Then Wesley College in Prahran said they had a drawer of old clarinet parts that had been sitting there since 1930. These inoperable instruments became the inspiration and foundation of *Escapee Gloss*, a seven-movement sound art composition.

The title *Escapee Gloss* is *A Symphony of Polymedia.* Polymedia is an unusual term and might be seen as a synonym for multimedia, so my use of the term needs explanation. It is, of course, like multimedia in that it refers to using several different media simultaneously. But, whereas multimedia expresses the simple fact of presence, polymedia, to my mind, expresses a sense of unity, too a sense that this mélange of media works towards a common goal and makes up a single coherent work of art. This is why I have also used the word symphony, which implies unity in diversity. Whether one takes the word *symphony* with the Greek translation (*symphonia*), or Mahler's idea about the symphony or that of Sibelius, a symphony is something that melds many diverse voices and many diverse elements into a unified whole. A polymedia symphonist does that and melds many other elements as well. Images, video, graphic scores, extended techniques, fragments of other works, even using the video as a moving score for improvisation, all working together, ideally, to express a single idea, or, better, to express the diversity and multiplicity of a single idea.

My practice is the performer, composer and visual artist creating an the identity of the whole, integrating acoustic sound, live audio mulching and acousmatic sound, all composed for reception via multiple loudspeakers, live video mixing and fixed video during performance.

2. DESCRIPTION

In the context of polymedia, *Escapee Gloss* transmutes the energy produced by music into a visual form, recognisable to the eye and exposing the inner necessity of the artist to explore the values both fragile and strong, common to both sonic and visual worlds. The different sensory worlds created in each movement are intended to encourage the listener to think about how music and visuals complement each other, and gain new strength from their combination.

Each movement of *Escapee Gloss* is named after a veneer of paint and explores avenues of texture, reflection, layers, light, colour, timbre, structure and the interaction of different media and acoustic sound and space. *Shine, Scintillating, Gloss, Pantone, Matte, Sheen, Silk.* Each name evokes a different thought or abstract meaning that I associate with the clarinet and manipulated electronic sounds. The musical/visual idea each veneer evokes is a different texture, literally, as each veneer has its own individual attributes.

Each movement of this composition explores a different disembodiment of the clarinet, from the plain

presentation of instrument pieces in *Sheen* and the glimpses of clarinet parts in *Silk* to the kaleidoscope of sound and visuals in *Pantone,* in which the clarinet, sound and visuals both is not recognizable. And, in *Gloss,* for a further extreme, the exploration has been extended into different sound worlds and visual layers using electronics and other instruments.

The main structural technique of this composition is layering—both in the audio and in the video—which quickly and efficiently create a variety of complex situations from a simple pile of unusable clarinets. These clarinets and clarinet parts have evoked many avenues, inspiring me to create sound worlds from the intimate to the extreme. This range is also projected in the visual from literally smashed up clarinets and glass to gorgeous bubbles, fine shadows and lines.

This series of compositions investigates the core of the music: tones, notation, instruments, composition, colour, design, line and texture. The following quote from Harry Partch contextualises how I have aimed to find a tangible meaning in *Escapee Gloss* through all the media presented, how I have taken these abandoned clarinets and transformed them into an exciting and sensitive polymedia work:

> The forms that imagination may devise transform the primitive sound-generation ideas into vehicles for new and exciting adventures, and the act of transforming in itself, like a fire by a stream, is an antidote to this age, a transcendence of its materials. And it is a small reaching back, through many thousands of years, to the first men who wished to find meaning for their lives through art.[1]

The title of each movement of *Escapee Gloss* is the name of a different texture of paint, and the bold colors of red, yellow and black are prominent in the videos. Silver, gold and white paint were used in different layers with the bold colours which transmuted many graduations of light continually transforming each piece of art or stills of the videos. Text prevails throughout this composition in all forms especially in *Silk,* to vocalize thoughts of how and why the music has come about. While the musical composition was usually written first, without the fragments of texts and the visuals of the disembodied clarinet underpinning the whole it is these that provide the concept.

The finished movements involved layering of different sounds and visuals, this layering of different textures in the sounds and light diffusions in the visuals created the pulse of the composition that drives the movement in Escapee Gloss. This keeps the composition moving both audibly and visually, so that no matter how complex or chaotic the combinations, the idea is never lost. The essence is to create a traveling moment with a start, middle and end, with many subtle and boisterous gestures and a sense of surprise and humour.

In performance, the musician plays Bb clarinet and controls the audio samples through the laptop and visual samples through the interactive responses on the video mixer, which is directly influenced by the sonic output of the system.

Gloss, Shine, and Silk involve other performers who are directed through notated/graphic/visual scores and cues, while *Pantone, Sheen* and *Matte* rely on the visual output screened in real time. During the performance, one is continually creating new musical ideas that trigger fresh sonic improvisations that then influence different combinations of visual layering that continually and fluidly change the balance between the visual and audio output. Both are reacting off each other at all times. The visual output is based on an abstract representation of how the music is progressing. The audio is affected by the live video feed, by the lighting and by the amount the performers interact with the footage in real-time. Being a polymedia symphonist is about integrating the sound and image structurally.

All audio processing during performance is created using Audio Mulch, a live audio interface for real-time audio performance.

The audio and visuals are all triggered manually. This is purely by choice, as I treat all the components individually in a polyphonic manner. The natural state for an instrumentalist is to have no attachments to the instrument; so, reaction to musical moments is spontaneous. However adding a laptop provides another line of the control that influences the outcome, resulting in a fluid and reactive performance.

The instrumentation, props and media included are the combination clarinet parts, added contraptions – microphones and cameras, multimedia visual works using parts of the clarinet and related woodwind instruments, improvisations, notated solo clarinet compositions and an electro-acoustic sound art installation. The audio works are inspired by and based on a series of short films.

Examples of electronic plug-ins I have used in the processing of the audio samples include delay, stereo delay, transposition, granulation, grain duration, pan and ring modulation. The process continues with manipulation of files into different layers and different channels, concentrating on microtonal interaction between the samples. A similar process is applied to the visual materials, including analysis of brightness, colour, contrast, duration, speed and complexity. The images

[1]Harry Partch, *Letter to William M. Bowen,* (New York: MS Lingua Press, 1967) 119.

have two categories: graphic-based images and film/still images. The sound and image influences the shape and analysis of each of the works. The audio in the compositions uses a real-time environment of acoustic sound and generative structures.

The imagery in *Escapee Gloss* is all about the dissolve and the transformation of the images. Very little manipulation is involved to fade one image to the next; only the length of the fade is calculated. The layering of the transparencies with luma, chroma, speed (pulse), and cutouts dominate many of the visual samples. The aim is to make the still images move through these effects through different diffusions of light.

3. ESCAPEE GLOSS: A SYMPHONY OF POLYMEDIA IN SEVEN MOVEMENTS

Escapee Gloss composed from 2010-14 will now be discussion of individual movements focussing on the first 3 movements *Shine, Scintillating and Silk* that use interactivity and layering of the audio and visuals with a short description of the last *Matte, Pantone. Sheen* and *Gloss.*

3.1.1. ***Movement 1 – Shine***

For Solo Bb Clarinet, electronics and Visuals

Shine explores a palette of extended clarinet techniques and acousmatic sound, focusing particularly on breath, white noise and timbre. The pitch organization is drawn from a twelve-tone spiral with a converging and diverging whole-tone scale. The clarinet part is also scored in graphic notation and live electronics that indicates the contour and duration for use of the scale effect and sample throughout the piece. The visual elements consist of delicate lines and bubbles crossing between greys, blues and sepia gradations. These intersect with the timbres of the clarinet part, electronic sounds and the angular contours of the musical phrases.

A pulse was created through the gentle movement of bubbles and lines both visually and audibly. My palette of extended clarinet techniques and flourishes focuses here particularly on breath and timbre. This, along with a virtuosic use of quartertones in melodic passages, both romantic and lyrical, aims to create an intensely personal, moving and transfixing bubble of fragility. In performance, one should use the acoustics of the space to create stillness and a sense of anticipation. There is a scene with the translucent surfaces and twisting lines traveling between the different layers.

The layering is replicated in the slow moving pulse of the sepia bubbles that prevails throughout, a kaleidoscope of graphics in sepia and blue gradients that at times dominates the space, appears to be a haze and an irregular pulse over the whole visual experience. The clarinet score and the video are a conversation, creating a layer of sensory completeness in the most intimate way.

The pitch organization of the score is drawn from a twelve-tone spiral with a converging and diverging whole-tone scale taken from Slonimsky's *Thesaurus*[1] *and Melodic Patterns*. The clarinet part is also scored in graphic notation that indicates the contours and durations for use of the scale throughout the piece.

Atmosphere is created through the pulse of the visuals created by subtle swirling with hollow sounds that are notated with big graphic notes in the score. These graphic notations are interpreted by the performer using their intuition and timing so they can take between half a second to four seconds according to the acoustics of the space.



Figure 1. Brigid Burke *Shine*

3.1.2. ***Movement 2 – Scintillating*** *For Electronics and visuals*

Scintillating was inspired by portraits of two clarinetists, one of myself and the other of my sister Grania. The portraits have been placed side by side as the opening visual in the film, as if in conversation (See Figure 2 & 3).

Scintillating began as a fully notated score for Bb clarinet and bass clarinet that was then transformed into an acousmatic composition through filters, fragmentation, rhythmic manipulations, layering and visual elements. The colors and texture are bold and simple as is the relentless pulse, two strong characters, forever voicing their intention, creating at times anguish and stimulation.

These dramatic figures were then transformed into a ghostly presence with intense black ink strokes to create, in the words of Clarence Barlow, "a metamorphosis of dancing objects."[2]

The opening is sombre and atmospheric, but this mood is very shortly interrupted by vibrant and energetic motifs. Syncopation and punctuations of repeated rhythms work against each other and timbral

[1] Nicolas Slonimsky, *Thesaurus of Scales and Melodic Patterns* scale, (New York: Charles Scribner's Sons 1947), 123.

[2]Clarence Barlow *Visualising Sound – Integrating sound and visual image as artform from Relationships Acts VIII des travaux 2004-2005 de l'Academie Internationale de Musique Eletroacoustique*, (Bourges: Mnemosyne musique, 2004/2005), 23.

alterations move in and out of sustained tones. These repeated rhythms make an amalgam of tonality and pulse that keeps the work in total harmony with itself.

Scintillating began as a fully notated score for Bb clarinet and bass clarinet that was then transformed into an acousmatic composition through filters, fragmentation, layering and visual elements. The materials that have been employed in the creation of the artworks associated with *Scintillating* are canvas, gold, bronze and silver leaf, resin, impasto, acrylic and house paint, silk screen prints, pen and ink and hand printed paper. The video footage used throughout takes as its starting point the two portraits, which includes flames seen through green luscious leaves. The colors used for the portraits are contrasting: red, black and silver in my portrait and yellow, gold and black for bass clarinet. These choices of colors and texture are bold and simple as is the relentless pulse, which operates in *Scintillating*.

The images grow smaller or switch all over the screen, depicting rhythms in the audio. The squares expand, merge with different fields of light. The images float as a field of space.

The video are a selection of parts of the portraits processed digitally, creating layered and refined artworks. The video takes the viewer on a journey with parts of the portraits emerging from each other to result in snapshots traveling over the screen. The original concept in the notated Bb and bass clarinet was to make it seem as though only one person was performing while in fact, the Bb clarinet and bass clarinet are in conversation, almost in unison at times, but as the piece progresses the separation evolves musically as well as visually.



Figure 2. Brigid Burke *Self Portrait*

Figure 3 Brigid Burke *Portrait of Grania*

3.1.3. ***Movement 3 – Silk*** *For Bass/Eb clarinet, bass clarinet/voice, flute/voice, prepared piano, electronics and visuals.*

Silk, is made up of random verbalizations about my art. *Silk* is dark and joyous with unexpected "circus" moments. It contextualizes thoughts into an elaborate score with vibrant interactions of different mediums of sound, performance and visuals.

Silk is a strong, soft and lustrous fabric of melodic lines and text combined with natural colours of red, blue and ochre. The interweaving of the melodic line, textures, timbres and refraction of light creates the feel of smooth textures. The silk screenings of text, clarinet parts and lines superimposed create this depth in the visuals and music as seen in Figure 4.

Silk is divided into two sections, connected by the electronics. Section one (duration 5'30") starts with metallic prepared piano chords followed by the two bass clarinets interplaying with each other with multiphonics and low single notes, with whistle tones from the flute growing in and out of each other. Repeated chords from the piano quickly interrupt the long ambient sounds. Then the action starts with these verbal utterances of fragments of text from both the electronics, flute and bass clarinet performer as they interpret the graphic notation. The sounds from the flautist are spoken into the mouthpiece creating a more intimate breathiness with accented starts and ends to the fragmentation of the words. The bass clarinet on the other hand is uttering words that come across as random moments with guttural utterances, gestures and individual notes with melodic gestural flourishes in the lower part of the instrument. The subtle melodic gestural flourishes of the second bass clarinet add to this excitement.

The extremity of registers from all the wind instruments and electronics with the bass clarinet changing to Eb clarinet at 3min18sec, to add to the structured chaos and extremity of sounds with the continuous metallic and distorted pulse from the piano.

The second section begins with sombre and playful electronic vocal sounds, a long section of almost solo melodic angular phrases from the bass clarinet that are punctuated at times by the fragmented piano rhythms and electronic outbursts.

The electronics are derived from voice utterance and these add another layer of suspense with the long sustained multiphonics, whistle tones and passages of melodic phrases in the bass clarinet. These are in conjunction with the extreme high, low and speech-like abrupt sounds produced from, speech, flute and clarinet sources. All sounds are processed, fragmented, manipulated, and mixed through computer software sound packages. The 'peaks' and subtle layers, and repeated notes in the piano ostinato passages of repeated rhythms and dynamically erratic clusters from

the wind instruments give the work an conversational feel.

The prepared piano motives and clarinet extended sounds which included voice and multiphonics were notated and recorded then divided into small samples to create the electronic sounds heard throughout. As the process of creating the score was recorded throughout the rehearsal process, the final score became audible and the video samples of both the still images and video clips were added. The refining the audio electronic music samples was similar to the rehearsal refinement of the notated score and this was done in the multi-channel program *Adobe Audition*.

The way the voice was used is one of the most interesting techniques used in the improvisational section. It adds another dimension, especially on the bass clarinet and flute with acoustic sounds and manipulated and random accented vowels that peak with unexpected utterances, as if each of the instruments are in conversation with each other. This improvisational technique adds to the overall timbre and movement of the work. Another technique used is singing higher melodies and playing lower notes. One hears the breaking down of sounds with fragments and imitations of other instruments which leaves the bass clarinet and flute sounding quite extreme. It can be heard especially in the bass clarinet line of *Sheen* and in *Silk* in which it is based around text that is accented with sampled vocal percussive sounds manipulated live electronically in the computer, prepared piano and verbal utterances from wind instruments using different vowels and vocal sounds while playing the wind instruments.

The text at times evolves into complex paths that are made up of an array of lines and shapes. This is achieved using simple plug-ins and effects which are reapplied to the image. The same process is used in the audio samples from the clarinets, piano and flute. Every still/moving image and audio sample is analysed so the visuals/audio shift, add and subtract from each other. The results are appealing patterns that emerge and continually evolve and grow.

The visual media also included wood, gold, bronze, silver leaf, resin, impasto, house paint, silk screen prints, pen and water color paper, pencil drawings, video footage of which the text was printed into and superimposed over to create these series of images. The superimposed multiple images create a flickering effect throughout *Silk.* The silkscreen prints ghostly register of fussiness suggests the difficulty of pinning down the text more than a traditional typeset of black and white text would have done. The role that these mediums play in communicating the expressive intent of *Silk* is expressed through this technique of layering the silk-screened prints of the text.

The acoustic sounds relate to how they have been digitally processed. The clarinet flutter tongues, sings while playing, speaks and whispers words. The percussive sounds from the prepared piano with repeated chords have been muted by paper inserted across the strings of the piano. The flute whispers, voice fragmentations and whistle tones prevail throughout. Thick woodwind textures with layering of virtuosic melodic and angular phrases are supported by the prevailing pulse from the prepared piano.

The acoustic samples are processed electronically and combined with the samples of the text, sound and sampled video projections. These visual components are then layered and manipulated in the computer. The words are manipulated by the instrumentalists using a notated score. They interpret them by accentuating certain parts of the vowels and words which are graphically pictured. See figure 4 of Page 6 of score.
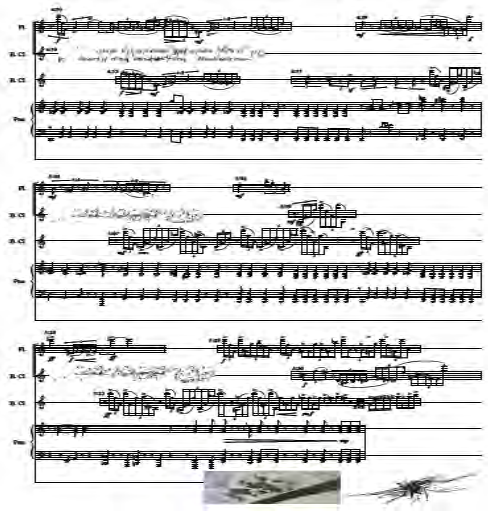


Figure 4. Brigid Burke *Silk* Excerpt from Score Page 6

Each component of sound and visual parts is based on the manipulation of rhythmic pulses created through the selected text that has been spoken, sung and played in different ways by the performers.

The sonic overtones of the woodwinds are explored in detail over the whole work, relating how they are digitally processed from the clarinet flutter tongues, singing while playing, verbal utterances and whispers along with percussive sounds from the prepared piano. The acoustic samples are processed electronically and combined with the pre-recorded samples of the text and sampled video projections. The disintegration of the clarinets is mirrored in the disintegration of text, heightened by the use of polymedia that enables communication on this complex sensory plane.

Timbre is a significant element in *Escapee Gloss,* and this is especially apparent in *Silk.* The use of

the prepared piano, where paper is layered on the strings inside the piano, creates quite a unique timbre character. Other timbral motifs in this piece include extended techniques and the melodic fanfare ideas.

The prepared piano creates a prevailing pulse with the repetitive quaver and crotchet chords. These chords were inspired by the repetitive sounds in the piano part of Malcolm Arnold's Sonatina. While Arnold's chords in Sonatina (as seen in figure 3.21) are consonant and form part of the harmonic structure, in *Silk* the chords have been assorted in no cohesive order and play no role within the harmonic structure. By replicating and distorting (due to the paper inserted into the piano) these chords, the timbral effect created is harsh, unresolved and dissonant. The chords are played as block chords or glissando chords and repeated throughout the piece.

The second timbral motif is the use of extended techniques, including multiphonics, overblowing and tongue clicks. However many of the extended techniques, such as voice utterances, hums, screams, whispers, whistle tones and throat sounds, stem from the use of the text. The text plays a significant role throughout and is derived from words and phrases including, "state of mind, a painting, free, chaos, colour, warmth, randomness, stillness, tranquility…" The performer may choose to make sounds using the text by sounding the vowel sounds, a single letter, utter the word or sound fragments of the word. These words or phrases sometimes appear within the graphic notation, improvised sections where the performer is able to draw on their palette of techniques including extended techniques.

Inspired by Arnold's fanfare flourishes in Sonatina and *Fantasy* for Clarinet[1] melodic flourishes appear in the wind parts of *Silk*, however these are employed more for their timbral qualities rather than melodic qualities. Fragments of flourishes appear throughout the piece although they are more prominent in the middle section. It is during this time that the texture thickens as the tutti sections become more frequent. Prior to this build up, the texture is not as thick as instrumental lines interweave with melodic, timbral and rhythmic motifs. These flourishes are disjunct, with complex rhythms. There are fragments of sequences within the motif that are repeated. For example at 4'33'' in the bass clarinet, the triplet ascending idea repeats three times in a short fragmented section. Another example is the ascending first half of the second group of sextuplets at 3'18'' (Bb, G#, E) and the descending second half (high E, A, A), repeated within the fragment at 3'40''.

[1] Malcolm Arnold, *Fantasy for Bb clarinet* (London: Alston & Co. Ltd., Amersham, Bucks, 1966).

The electronic part is a mix of sombre and playful outbursts derived from samples of the text from the voice utterances, and from the percussive piano sounds. The opening begins with a low pulsing drone motif that continues and is fragmented throughout the work. Metallic motifs enter throughout as short fragments or longer passages and the distorted sounds of the prepared piano chords are evident in the second section.



Figure 5. Brigid Burke *Silk*

*3.1.4. Movement **4** – Pantone For Bass clarinet, double bass, electronics and visuals.*

The finished video is the actual score. The performers react to the moving images to create the audio. This is an interactive electronic audio performance that includes spatialized sound and live visual elements. The images of *Pantone* are an ever-dynamic kaleidoscope, interrupted by gentle water moving through the seascape of the glass plate as seen in figure 5. The audio from the bass clarinet, double bass and sampled water electronics makes a subtle counter-point that punctuates and accentuates these delicate moving images, creating depth and interesting nuances.



Figure 6. Brigid Burke *Glass plate with kaleidoscope*

*3.1.5. **Movement 5 – Matte** For Visuals and electronics.*

Matte was inspired by the soft environmental soundscapes of rainforests. The clarinet produces wind sounds throughout. The clicking of clarinet keys, percussive cymbal and bowed vibraphone metallic sounds mimic the sounds of rain. This was then visualized in the artwork of *Matte* using round pieces of coloured glass, small broken clarinet parts and both wooden and silver keys, which were moved through

water in a glass bowl. These images were combined with a series of blue mixed-media works and coloured pencil drawings on paper formed in the shapes of droplets, a series of squiggle-based mixed media works using acrylic paint, pen and ink and charcoal works on paper which were then photographed as seen in Figure 6.



Figure 7. Brigid Burke *Matte,*

***3.1.6. Movement** 6 – Sheen For Bass clarinet, live electronics/laptop and live feed and pre-recorded video projection.*

Sheen is an improvised quartet for clarinet and real-time audio mulching with live video feed and prerecorded video. *Sheen* has an intrinsic glistening quality and the keys of the clarinet are the focal point for this shining and resplendent movement.

Microphones and a laptop are used to control the synthesis of acoustic material sampled during the performance. The live clarinet interacts with these visual components: a camera filming a bowl of water filled with clarinet parts, another one filming the fingers and keys of the bass clarinet and a prerecorded version of clarinet parts on video that has already been processed. The bubbles and water movement are controlled by an aquatic pump. A pulse is created through such effects as strobes and finger movements from the prerecorded image and live feed of the bass clarinet.

All audio processing is done with Audio Mulch, including the live acoustic bass clarinet manipulated during performance.



Figure 8. Brigid Burke *Sheen, Bent Clarinet*

*3.1.7. **Movement 7 – Gloss – a series of 9 short connected movements.** For Electronics, Bb clarinet/bass Clarinet, Eb clarinet/bass clarinet, flute/pic, piano, double bass and visuals.*

Gloss was inspired by a series of nine pen and ink drawings, which were used as graphic notation for the different instrumental combinations and live electronics. The nine graphics were then integrated into the video art work which shows glimpses of the graphics in superimposed layers in conjunction with pencil drawings of squiggles depicting wire as seen in Figure 8 These were then photographed. The depiction of wire glosses the image with a superficial lustre.



Figure 9. Brigid Burke *Gloss 6*

4. CONCLUSION

This paper has examined one major composition, *Escapee Gloss that* represents a map of artistic development and a window into my ongoing artistic practice. As a clarinet performer, I have always enjoyed the immediacy of live performance and the direct engagement with an audience. While Escapee Gloss, I could see infinite possibilities offered through the application of processing to both recorded and live sound.

5. REFERENCES

Books

Partch, Harry. *Genesis of a Music: An Account of a creative work, its roots and its fulfillments.* London: Da Capo, 1949.

Read, Gardener. *Breath and Air sounds: Compendium of Modern Instrumental Techniques.* Connecticut: Greenwood Press, 1993.

Reynolds, Roger. *Notations by John Cage*. New York: Something Else Press, Inc., 1969.

Slonimsky, Nicolas. *Thesaurus of Scales and Melodic Patterns*. New York: Charles Scibner's Sons, 1947.

Weibel, Peter and Gregor Hansen. 2006. "The Development of Light VII From Synesthesia. Synthetic Sounds and Images". in *Light Art from Artificial Light, Light As a Medium in the Art of the 20th And 21st*

Centuries. Hatje Cantz Pub Germany: Hatje Cantz Verlag.

Stockhausen, Karlheinz. 2009. *Cosmic Pulses. Notations 21* Theresa Saucer. New York: Notations 21 Mark Batty.

Xenakis, Iannis. 2010. "How Do You Draw Sound?". *Polytope de Montreal*, edited by, Carey Lovelace,& Sharon Kanach New York: Drawing Centre.

Proceedings paper

Bisig, Daniel and Tatsuo Unemi. 2009 "Swarm Simulations for Dance Performance". *Twelfth Generative Art Conference*, Institute of Computer Music and Sound Technology Zurich University of Arts, Zurich.

Errante, F. Gerard. "The Electric Clarinet, Part I Repertoire." *ClariNetwork* 32, No.2 (2005)
"The Electric Clarinet, Part II Repertoire." *ClariNetwork* 32, No.3 (2005).
"Electro-Acoustic Music for the Clarinet- Part 1" ClariNetwork (Fall 1985).
"The Electric Clarinet Part 1" The Clarinet 32, No 2 (March 2005).
"*The Electric Clarinet Part 11"* The Clarinet 32, No 3 (June 2005).

Kim-Boyle, David. "Extended Non-Linear Applications and Visual Aesthetics of Real-Time Scores". *Proceedings of the International Computer Music Association and the Western Australian Academy of Performing Arts*, Sydney, 2013.